TM Forum Specification

CDR Transaction Management

TMF735

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Introduction

The following document is the user guide of the REST API for CDR transaction management. It includes the model definition as well as all available operations.

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Sample Use Cases

Reader will find example of use cases using CDR Transaction API in “Zero-Touch Partnering Use-Cases” document.

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Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities, for instance a BillingAccount and SettlementAccount inheriting properties from the abstract Account entity.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines, Part 2 (TMF630).

The @type attribute provides a way to represent the actual class type of an entity. For example, within a list of Account instances some may be instances of BillingAccount where other could be instances of SettlementAccount. The @type gives this information. All resources and sub-resources of this API have a @type attributes that can be provided when this is useful.

The @referredType can be used within reference entities (like for instance an AccountRef object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the @type, when used, denotes the class type of the reference itself, such as BillingAccountRef or SettlementAccountRef, and not the class type of the referred object. However since reference classes are rarely sub-classed, @type is generally not useful in reference objects.

The @schemaLocation property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected characteristics of an entity.

The @baseType attribute gives a way to provide explicitly the base of class of a given resource that has been extended.

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RESOURCE MODEL

Managed Entity and Task Resource Models

CdrTransaction resource

A party revenue sharing transaction is a type of transaction which can be used to launch a revenue share between a partner and a service provider or between multiple partners.

Resource model

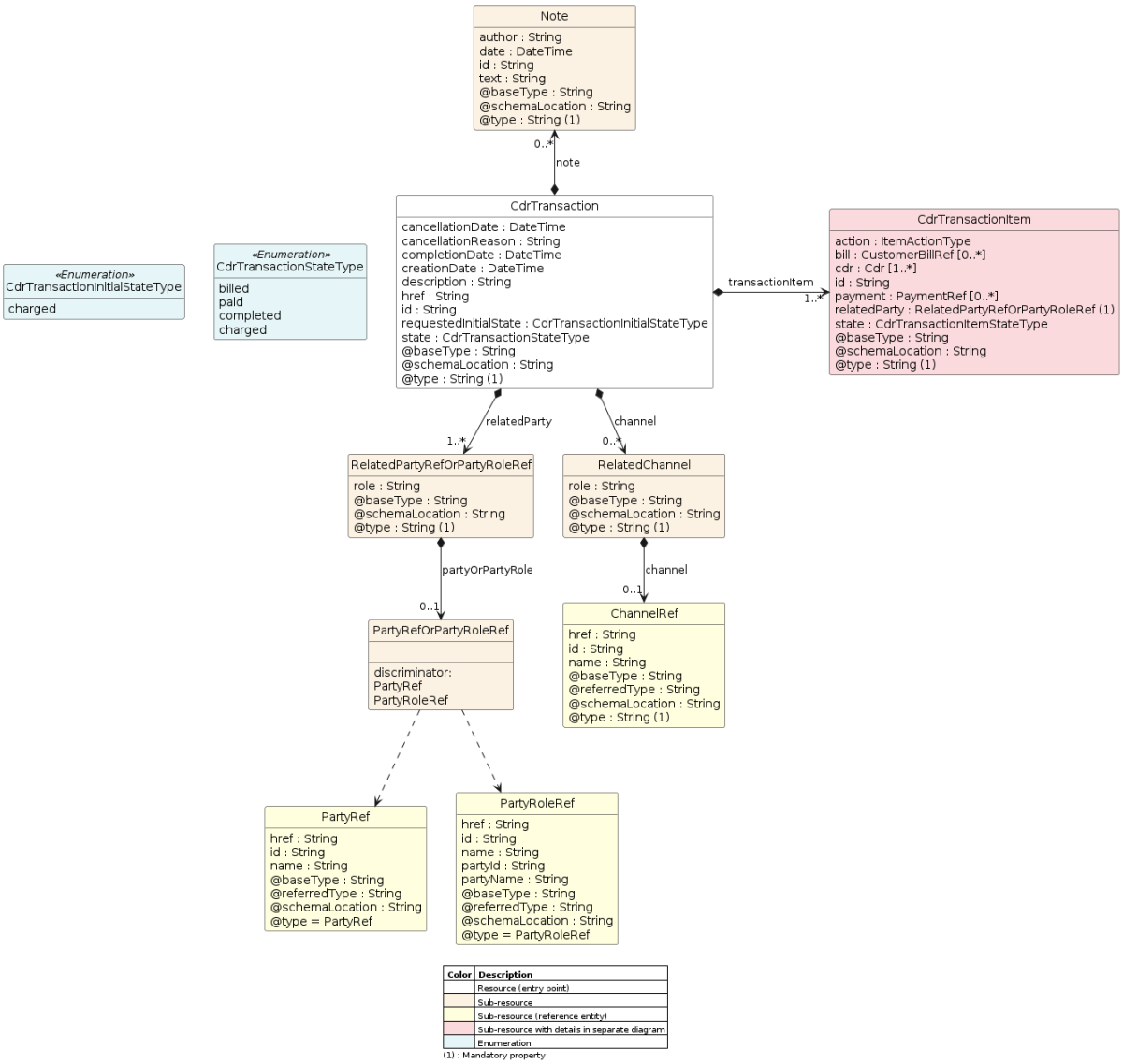


Figure 1 - CdrTransaction

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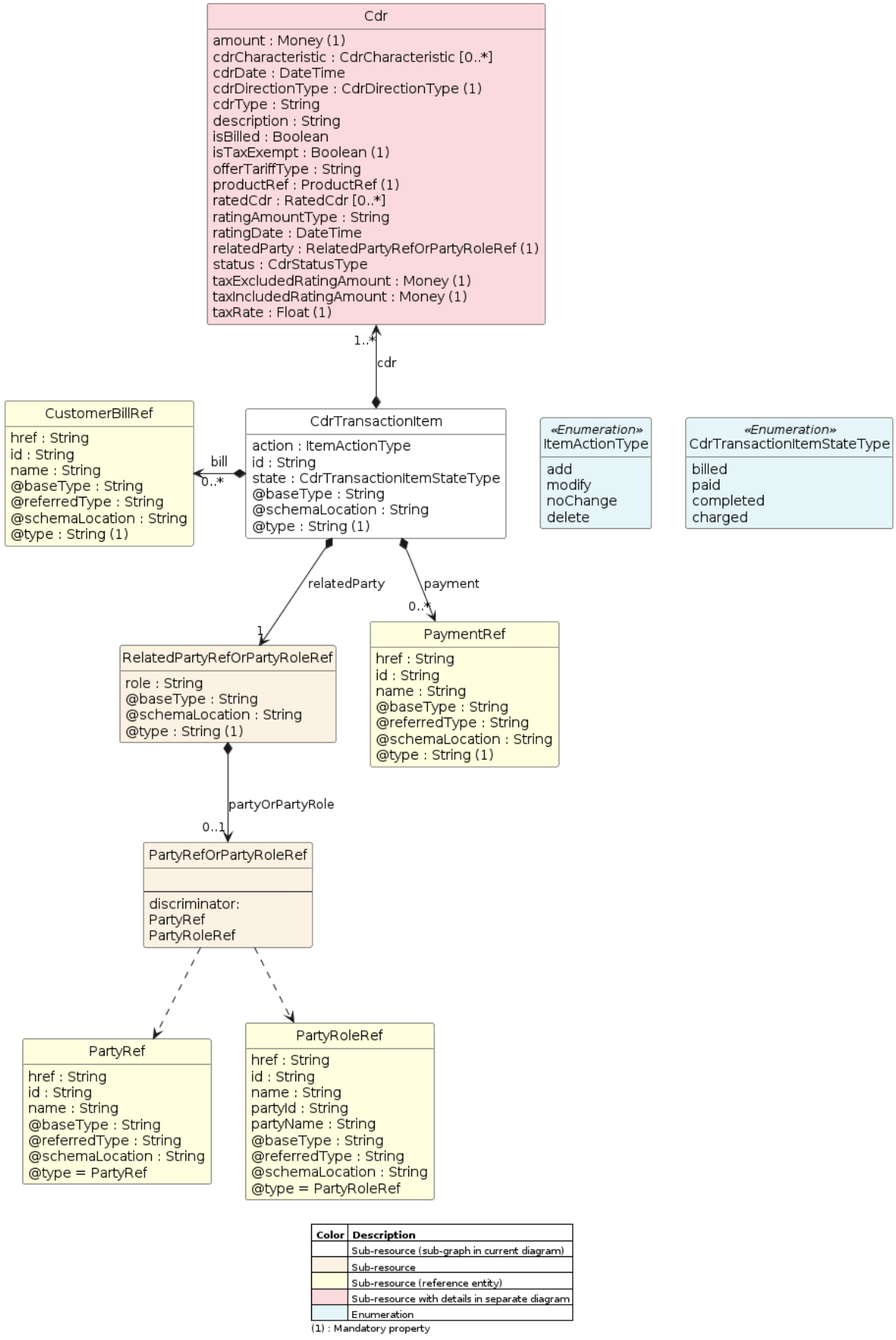


Figure 2 - CdrTransactionItem

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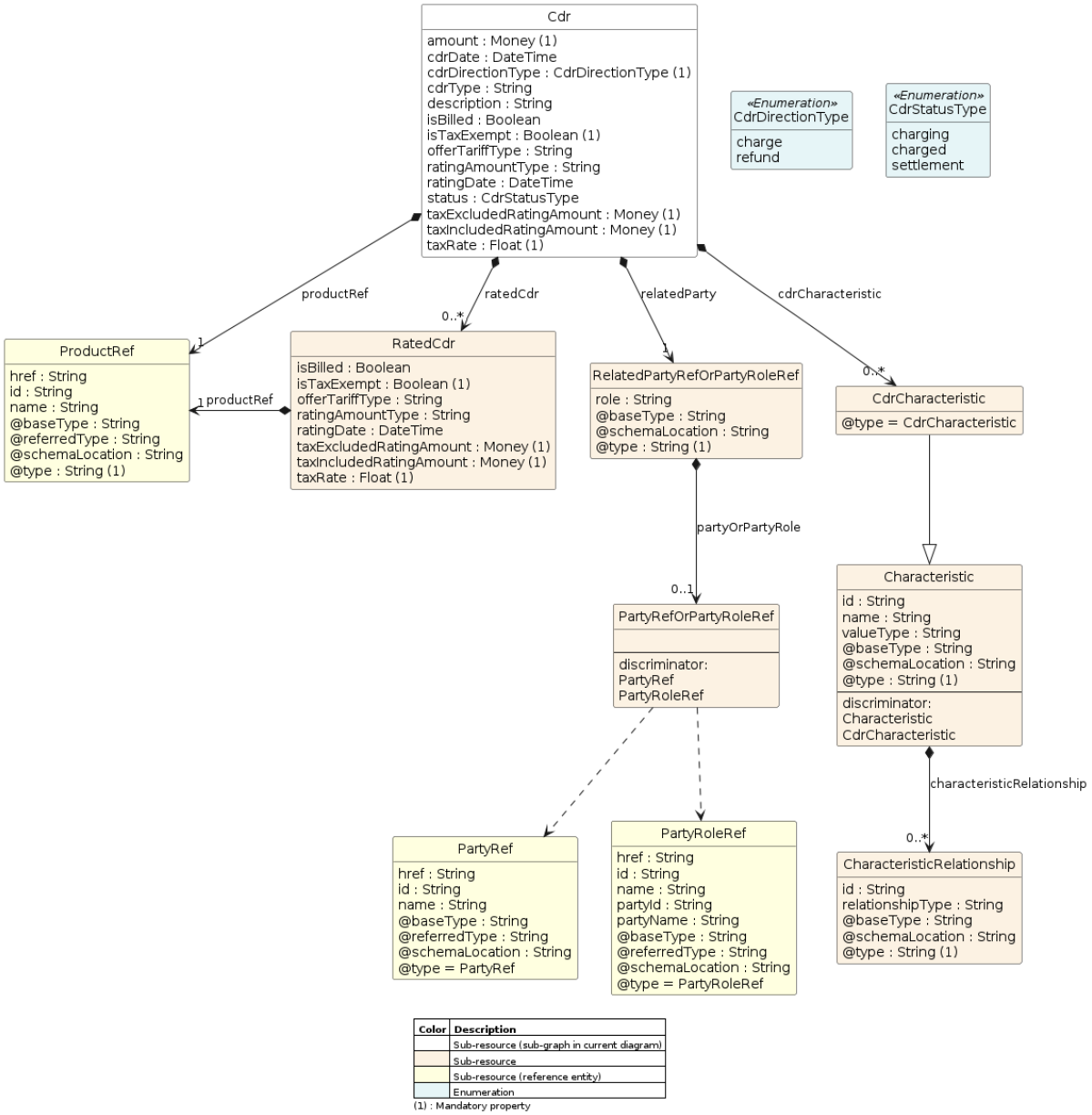


Figure 3 - Cdr

Field descriptions

CdrTransaction fields

cancellationDate	A DateTime. Date when the transaction is cancelled. This is used when transaction is cancelled.
cancellationReason	A String. Reason why the transaction is cancelled. This is used when transaction is cancelled.
channel	A RelatedChannel. Related channel to another entity. May be online web, mobile app, social ,etc.
completionDate	A DateTime. Date when the ProductOrder was completed.
creationDate	A DateTime. Date and time when the ProductOrder was created.

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description	A String. Description of the transaction.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
note	A Note. Extra information about a given entity.
relatedParty	A RelatedPartyRefOrPartyRoleRef. The associated party or partyRole.
requestedInitialState	A CdrTransactionInitialStateType. Possible values for the requested initial state of the transaction from client- by default created is considered.
state	A CdrTransactionStateType. Possible values for the state of the Cdr transaction.
transactionItem	A CdrTransactionItem. An identified part of the transaction. A transaction is decomposed into one or more transaction items.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Cdr sub-resource fields

amount	A Money. A base / value business entity used to represent money.
cdrCharacteristic	A CdrCharacteristic. Provides the value of a given characteristic.
cdrDate	A DateTime. Date of cdr.
cdrDirectionType	A CdrDirectionType. Possible values for the Direction of the Cdr.
cdrType	A String. Type of cdr.
description	A String. Description of cdr.
isBilled	A Boolean. Boolean indicating if usage have been billed or not.
isTaxExempt	A Boolean. Indicates if the rated amount is exempt of tax.
offerTariffType	A String. Type of tariff applied.
productRef	A ProductRef. A Product reference.
ratedCdr	A RatedCdr. An occurrence of employing a cdr for its intended purpose with all rating details.
ratingAmountType	A String. Type of amount.
ratingDate	A DateTime. Date of usage rating.
relatedParty	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.
status	A CdrStatusType. Possible values for the status of the Cdr.

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taxExcludedRatingAmount	A Money. A base / value business entity used to represent money.
taxIncludedRatingAmount	A Money. A base / value business entity used to represent money.
taxRate	A Float. Tax rate.

CdrTransactionItem sub-resource fields

action	An ItemActionType. Action to be performed on the entity managed by the item.
bill	A CustomerBillRef. The bills related to this report item.
cdr	A Cdr. An occurrence of employing a product for its intended purpose with all rating details.
id	A String. Identifier of the transaction item (generally it is a sequence number 01, 02, 03, …).
payment	A PaymentRef. The payments related to this report item.
relatedParty	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.
state	A CdrTransactionItemStateType. Possible values for the state of the party revenue sharing transaction.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ChannelRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Characteristic sub-resource fields

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characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CharacteristicRelationship sub-resource fields

id	A String. Unique identifier of the characteristic.
relationshipType	A String. The type of relationship.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CustomerBillRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Note sub-resource fields

author	A String. Author of the note.
date	A DateTime. Date of the note.
id	A String. Identifier of the note within its containing entity.
text	A String. Text of the note.
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRoleRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
partyId	A String. The identifier of the engaged party that is linked to the PartyRole object.
partyName	A String. The name of the engaged party that is linked to the PartyRole object.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PaymentRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.

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@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ProductRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RatedCdr sub-resource fields

isBilled	A Boolean. Boolean indicating if usage have been billed or not.
isTaxExempt	A Boolean. Indicates if the rated amount is exempt of tax.
offerTariffType	A String. Type of tariff applied.
productRef	A ProductRef. A Product reference.
ratingAmountType	A String. Type of amount.
ratingDate	A DateTime. Date of usage rating.
taxExcludedRatingAmount	A Money. A base / value business entity used to represent money.
taxIncludedRatingAmount	A Money. A base / value business entity used to represent money.
taxRate	A Float. Tax rate.

RelatedChannel sub-resource fields

channel	A ChannelRef. The channel to which the resource reference to. e.g. channel for selling product offerings, channel for opening a trouble ticket etc..
role	A String.
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedPartyRefOrPartyRoleRef sub-resource fields

partyOrPartyRole	A PartyRefOrPartyRoleRef.
role	A String. Role played by the related party or party role in the context of the specific entity it is linked to. Such as 'initiator', 'customer', 'salesAgent', 'user'.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'CdrTransaction' resource object.


```
{
  "transactionItem": [
    {
      "cdr": [{
        "taxRate": 0,
        "isBilled": true,
        "amount": {
          "unit": "USD",
          "value": 50000
        },
        "cdrType": "revenue share",
        "productRef": {
          "@referredType": "Product",
          "@type": "ProductRef",
          "name": "Mobile handset 2023 product of MyFirstPartner",
          "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7810",
          "id": "7810"
        },
        "description": "Mobile handset 2023 distribution price",
        "cdrDate": "2020-09-21T09:13:16-07:00",
        "id": "181011",
        "taxExcludedRatingAmount": {
          "unit": "USD",
          "value": 0
        },
        "relatedParty": {
          "partyOrPartyRole": {
            "@referredType": "Organization",
            "@type": "PartyRef",
            "name": "MyFirstPartner",
            "href": "https://mycsp.com:8080/tmf-
```


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```
api/partyManagement/v5/organization/2777",
            "id": "2777"
          },
          "role": "distributor",
          "@type": "RelatedPartyRefOrPartyRoleRef"
        },
        "taxIncludedRatingAmount": {
          "unit": "USD",
          "value": 0
        },
        "status": "charged"
      }],
      "@type": "CdrTransactionItem",
      "action": "add",
      "bill": [{
        "@referredType": "CustomerBill",
        "@type": "CustomerBillRef",
        "name": "Customer 202302 bill of MyFirstPartner",
        "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2001",
        "id": "2001"
      }],
      "payment": [{
        "@referredType": "Payment",
        "@type": "PaymentRef",
        "name": "Customer 202302 bill payment of MyFirstPartner",
        "href": "https://mycsp.com:8080/tmf-api/CustomerManagement/v5/Payment/5001",
        "id": "5001"
      }],
      "state": "billed",
      "id": "18101",
      "relatedParty": {
        "partyOrPartyRole": {
          "@referredType": "Organization",
          "@type": "PartyRef",
          "name": "MyFirstPartner",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
          "id": "2777"
        },
        "role": "distributor",
        "@type": "RelatedPartyRefOrPartyRoleRef"
      }
    },
    {
      "cdr": [{
        "taxRate": 0,
        "isBilled": true,
        "amount": {
          "unit": "USD",
          "value": 30000
        },
        "cdrType": "revenue share",
        "productRef": {
          "@referredType": "Product",
          "@type": "ProductRef",
          "name": "Smart speakers 2023 product of MyFirstPartner",
          "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7811",
          "id": "7811"
        },
        "description": "smart speakers 2023 distribution price",
        "cdrDate": "2020-09-21T09:13:16-07:00",
```


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```
        "id": "181021",
        "taxExcludedRatingAmount": {
          "unit": "USD",
          "value": 0
        },
        "relatedParty": {
          "partyOrPartyRole": {
            "@referredType": "Organization",
            "@type": "PartyRef",
            "name": "MyFirstPartner",
            "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
            "id": "2777"
          },
          "role": "distributor",
          "@type": "RelatedPartyRefOrPartyRoleRef"
        },
        "taxIncludedRatingAmount": {
          "unit": "USD",
          "value": 0
        },
        "status": "charged"
      }],
      "@type": "CdrTransactionItem",
      "action": "add",
      "bill": [{
        "@referredType": "CustomerBill",
        "@type": "CustomerBillRef",
        "name": "Customer 202302 bill of MyFirstPartner",
        "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2001",
        "id": "2001"
      }],
      "payment": [{
        "@referredType": "Payment",
        "@type": "PaymentRef",
        "name": "Customer 202302 bill payment of MyFirstPartner",
        "href": "https://mycsp.com:8080/tmf-api/CustomerManagement/v5/Payment/5001",
        "id": "5001"
      }],
      "state": "billed",
      "id": "18102",
      "relatedParty": {
        "partyOrPartyRole": {
          "@referredType": "Organization",
          "@type": "PartyRef",
          "name": "MyFirstPartner",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
          "id": "2777"
        },
        "role": "distributor",
        "@type": "RelatedPartyRefOrPartyRoleRef"
      }
    },
    {
      "cdr": [{
        "taxRate": 0,
        "isBilled": true,
        "amount": {
          "unit": "USD",
          "value": 60000
        },
```


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```
        "cdrType": "revenue share",
        "productRef": {
          "@referredType": "Product",
          "@type": "ProductRef",
          "name": "Mobile handset 2023 product of MyFirstPartner",
          "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7810",
          "id": "7810"
        },
        "description": "Mobile handset 2023 distribution price",
        "cdrDate": "2020-09-21T09:13:16-07:00",
        "id": "181031",
        "taxExcludedRatingAmount": {
          "unit": "USD",
          "value": 0
        },
        "relatedParty": {
          "partyOrPartyRole": {
            "@referredType": "Organization",
            "@type": "PartyRef",
            "name": "Mobi Service Providor",
            "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2779",
            "id": "2779"
          },
          "role": "revenueSharingModelOwner",
          "@type": "RelatedPartyRefOrPartyRoleRef"
        },
        "taxIncludedRatingAmount": {
          "unit": "USD",
          "value": 0
        },
        "status": "charged"
      }],
      "@type": "CdrTransactionItem",
      "action": "add",
      "bill": [{
        "@referredType": "CustomerBill",
        "@type": "CustomerBillRef",
        "name": "Customer 202302 bill of MyFirstPartner",
        "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2002",
        "id": "2002"
      }],
      "state": "billed",
      "id": "18103",
      "relatedParty": {
        "partyOrPartyRole": {
          "@referredType": "Organization",
          "@type": "PartyRef",
          "name": "MyFirstPartner",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
          "id": "2777"
        },
        "role": "distributor",
        "@type": "RelatedPartyRefOrPartyRoleRef"
      }
    }
  ],
  "requestedInitialState": "charged",
  "@type": "CdrTransaction",
  "channel": [{
```


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```
    "role": "revenueSharingDepartment",
    "@type": "RelatedChannel",
    "channel": {
      "@referredType": "Channel",
      "@type": "ChannelRef",
      "name": "revenue sharing department",
      "href": "https://mycsp.com:8080/tmf-api/channelManagement/v5/channel/1777",
      "id": "1777"
    }
  }],
  "description": "the transaction of MyFirstPartner revenue share",
  "completionDate": "2023-03-10T07:37:23.112Z",
  "href": "https://mycsp.com:8080/tmf-
api/CdrTransactionManagement/v5/CdrTransaction/1810",
  "id": "1810",
  "state": "charged",
  "relatedParty": [
    {
      "partyOrPartyRole": {
        "@referredType": "Organization",
        "@type": "PartyRef",
        "name": "MyFirstPartner",
        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
        "id": "2777"
      },
      "role": "distributor",
      "@type": "RelatedPartyRefOrPartyRoleRef"
    },
    {
      "partyOrPartyRole": {
        "@referredType": "Organization",
        "@type": "PartyRef",
        "name": "Mobi Service Providor",
        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2779",
        "id": "2779"
      },
      "role": "revenueSharingModelOwner",
      "@type": "RelatedPartyRefOrPartyRoleRef"
    }
  ],
  "creationDate": "2023-03-10T07:37:23.112Z"
}
```


Notification Resource Models

4 notifications are defined for this API.

Notifications related to CdrTransaction:

- Cdr Transaction Delete Event
- Cdr Transaction Attribute Value Change Event
- Cdr Transaction Create Event
- Cdr Transaction State Change Event

The notification structure for all notifications in this API follow the pattern depicted by the

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figure below. A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType). This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

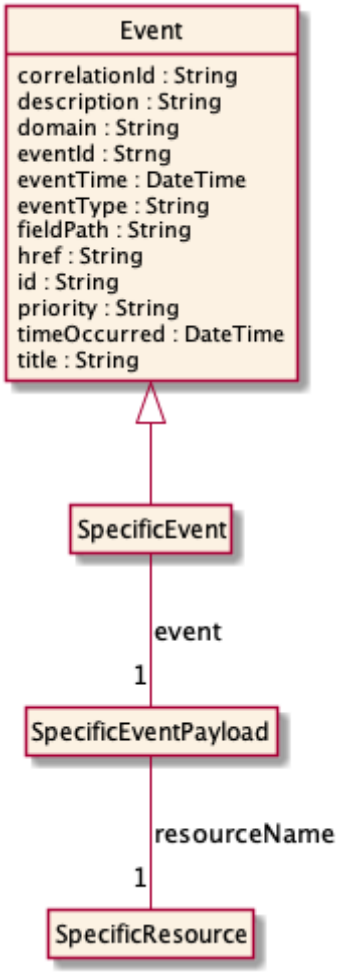


Figure 4 Notification Pattern

Cdr Transaction Delete Event

Message example for CdrTransactionDeleteEvent event

```
Content-Type: application/json

{
    "correlationId": "93cfc73d-deb7",
    "description": "CdrTransactionDeleteEvent illustration",
    "domain": "BusinessPartner",
    "eventId": "48d6-9751-40e4f01440c9",
    "eventTime": "2022-08-25T12:18:12.202Z",
    "eventType": "CdrTransactionDeleteEvent",
    "priority": "4",
    "timeOcurred": "2022-08-25T12:18:07.224Z",
    "title": "CdrTransactionDeleteEvent",
```

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```
    "event": {
        "cDRTransaction": {
            "id": "1810",
            "href": "https://host:port/cDRTransaction/v5/cDRTransaction/1810",
            "@type": "CdrTransaction"
        }
    },
    "reportingSystem": {
        "id": "759",
        "name": "APP-745",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "705",
        "name": "APP-317",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "CdrTransactionDeleteEvent"
}
```


Cdr Transaction Attribute Value Change Event

Message example for CdrTransactionAttributeValueChangeEvent event


```
Content-Type: application/json

{
    "correlationId": "91cfc73d-deb7",
    "description": "CdrTransactionAttributeValueChangeEvent illustration",
    "domain": "BusinessPartner",
    "eventId": "57d6-9751-40e4f01440c9",
    "eventTime": "2022-08-25T12:18:12.202Z",
    "eventType": "CdrTransactionAttributeValueChangeEvent",
    "priority": "4",
    "timeOcurred": "2022-08-25T12:18:07.224Z",
    "title": "CdrTransactionAttributeValueChangeEvent",
    "event": {
        "cDRTransaction": {
            "@type": "CdrTransaction",
            "href": "https://mycsp.com:8080/tmf-
api/CdrTransactionManagement/v5/CdrTransaction/1810",
            "id": "1810",
            "relatedParty": [
                {
                    "@type": "RelatedPartyRefOrPartyRoleRef",
                    "role": "distributor",
                    "partyOrPartyRole": {
                        "@type": "PartyRef",
                        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
                        "id": "2777",
                        "name": "MyFirstPartner",
                        "@referredType": "Organization"
                    }
                },
                {
                    "@type": "RelatedPartyRefOrPartyRoleRef",
```


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```
                    "role": "revenueSharingModelOwner",
                    "partyOrPartyRole": {
                        "@type": "PartyRef",
                        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2779",
                        "id": "2779",
                        "name": "Mobi Service Providor",
                        "@referredType": "Organization"
                    }
                }
            ],
            "state": "charged",
            "requestedInitialState": "charged",
            "channel": [
                {
                    "@type": "RelatedChannel",
                    "role": "revenueSharingDepartment",
                    "channel": {
                        "@type": "ChannelRef",
                        "href": "https://mycsp.com:8080/tmf-
api/channelManagement/v5/channel/1777",
                        "id": "1777",
                        "name": "revenue sharing department",
                        "@referredType": "Channel"
                    }
                }
            ],
            "description": "the transaction of MyFirstPartner revenue share",
            "transactionItem": [
                {
                    "@type": "CdrTransactionItem",
                    "state": "billed",
                    "id": "18101",
                    "action": "add",
                    "cdr": [
                        {
                            "cdrDate": "2020-09-21T09:13:16-07:00",
                            "description": "Mobile handset 2023 distribution price",
                            "status": "charged",
                            "id": "181011",
                            "cdrType": "revenue share",
                            "relatedParty": {
                                "@type": "RelatedPartyRefOrPartyRoleRef",
                                "role": "distributor",
                                "partyOrPartyRole": {
                                    "@type": "PartyRef",
                                    "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
                                    "id": "2777",
                                    "name": "MyFirstPartner",
                                    "@referredType": "Organization"
                                }
                            },
                            "isBilled": true,
                            "isTaxExempt": false,
                            "productRef": {
                                "@type": "ProductRef",
                                "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7810",
                                "id": "7810",
                                "name": "Mobile handset 2023 product of
MyFirstPartner",
                                "@referredType": "Product"
```


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```
                            },
                            "taxExcludedRatingAmount": {
                                "unit": "USD",
                                "value": 0
                            },
                            "taxIncludedRatingAmount": {
                                "unit": "USD",
                                "value": 0
                            },
                            "taxRate": 0,
                            "amount": {
                                "unit": "USD",
                                "value": 50000
                            }
                        }
                    ],
                    "relatedParty": {
                        "@type": "RelatedPartyRefOrPartyRoleRef",
                        "role": "distributor",
                        "partyOrPartyRole": {
                            "@type": "PartyRef",
                            "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
                            "id": "2777",
                            "name": "MyFirstPartner",
                            "@referredType": "Organization"
                        }
                    },
                    "bill": [
                        {
                            "@type": "CustomerBillRef",
                            "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2001",
                            "id": "2001",
                            "name": "Customer 202302 bill of MyFirstPartner",
                            "@referredType": "CustomerBill"
                        }
                    ],
                    "payment": [
                        {
                            "@type": "PaymentRef",
                            "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/Payment/5001",
                            "id": "5001",
                            "name": "Customer 202302 bill payment of
MyFirstPartner",
                            "@referredType": "Payment"
                        }
                    ]
                },
                {
                    "@type": "CdrTransactionItem",
                    "state": "billed",
                    "id": "18102",
                    "action": "add",
                    "cdr": [
                        {
                            "cdrDate": "2020-09-21T09:13:16-07:00",
                            "description": "smart speakers 2023 distribution price",
                            "status": "charged",
                            "id": "181021",
                            "cdrType": "revenue share",
                            "relatedParty": {
```


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```
                                "@type": "RelatedPartyRefOrPartyRoleRef",
                                "role": "distributor",
                                "partyOrPartyRole": {
                                    "@type": "PartyRef",
                                    "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
                                    "id": "2777",
                                    "name": "MyFirstPartner",
                                    "@referredType": "Organization"
                                }
                            },
                            "isBilled": true,
                            "isTaxExempt": false,
                            "productRef": {
                                "@type": "ProductRef",
                                "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7811",
                                "id": "7811",
                                "name": "Smart speakers 2023 product of
MyFirstPartner",
                                "@referredType": "Product"
                            },
                            "taxExcludedRatingAmount": {
                                "unit": "USD",
                                "value": 0
                            },
                            "taxIncludedRatingAmount": {
                                "unit": "USD",
                                "value": 0
                            },
                            "taxRate": 0,
                            "amount": {
                                "unit": "USD",
                                "value": 30000
                            }
                        }
                    ],
                    "relatedParty": {
                        "@type": "RelatedPartyRefOrPartyRoleRef",
                        "role": "distributor",
                        "partyOrPartyRole": {
                            "@type": "PartyRef",
                            "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
                            "id": "2777",
                            "name": "MyFirstPartner",
                            "@referredType": "Organization"
                        }
                    },
                    "bill": [
                        {
                            "@type": "CustomerBillRef",
                            "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2001",
                            "id": "2001",
                            "name": "Customer 202302 bill of MyFirstPartner",
                            "@referredType": "CustomerBill"
                        }
                    ],
                    "payment": [
                        {
                            "@type": "PaymentRef",
                            "href": "https://mycsp.com:8080/tmf-
```


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```
api/CustomerManagement/v5/Payment/5001",
                            "id": "5001",
                            "name": "Customer 202302 bill payment of
MyFirstPartner",
                            "@referredType": "Payment"
                        }
                    ]
                },
                {
                    "@type": "CdrTransactionItem",
                    "state": "billed",
                    "id": "18103",
                    "action": "add",
                    "cdr": [
                        {
                            "cdrDate": "2020-09-21T09:13:16-07:00",
                            "description": "Mobile handset 2023 distribution price",
                            "status": "charged",
                            "id": "181031",
                            "cdrType": "revenue share",
                            "relatedParty": {
                                "@type": "RelatedPartyRefOrPartyRoleRef",
                                "role": "revenueSharingModelOwner",
                                "partyOrPartyRole": {
                                    "@type": "PartyRef",
                                    "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2779",
                                    "id": "2779",
                                    "name": "Mobi Service Providor",
                                    "@referredType": "Organization"
                                }
                            },
                            "isBilled": true,
                            "isTaxExempt": false,
                            "productRef": {
                                "@type": "ProductRef",
                                "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7810",
                                "id": "7810",
                                "name": "Mobile handset 2023 product of
MyFirstPartner",
                                "@referredType": "Product"
                            },
                            "taxExcludedRatingAmount": {
                                "unit": "USD",
                                "value": 0
                            },
                            "taxIncludedRatingAmount": {
                                "unit": "USD",
                                "value": 0
                            },
                            "taxRate": 0,
                            "amount": {
                                "unit": "USD",
                                "value": 60000
                            }
                        }
                    ],
                    "relatedParty": {
                        "@type": "RelatedPartyRefOrPartyRoleRef",
                        "role": "distributor",
                        "partyOrPartyRole": {
                            "@type": "PartyRef",
```


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```
                            "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
                            "id": "2777",
                            "name": "MyFirstPartner",
                            "@referredType": "Organization"
                            }
                    },
                    "bill": [
                        {
                            "@type": "CustomerBillRef",
                            "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2002",
                            "id": "2002",
                            "name": "Customer 202302 bill of MyFirstPartner",
                            "@referredType": "CustomerBill"
                        }
                    ]
                }
            ],
            "creationDate": "2023-03-10T07:37:23.112Z",
            "completionDate": "2023-03-10T07:37:23.112Z"
        }
    },
    "reportingSystem": {
        "id": "759",
        "name": "APP-745",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "705",
        "name": "APP-317",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "CdrTransactionAttributeValueChangeEvent"
}
```


Cdr Transaction Create Event

Message example for CdrTransactionCreateEvent event


```
Content-Type: application/json

{
    "correlationId": "92cfc73d-deb7",
    "description": "CdrTransactionDeleteEvent illustration",
    "domain": "BusinessPartner",
    "eventId": "67d6-9751-40e4f01440c9",
    "eventTime": "2022-08-25T12:18:12.202Z",
    "eventType": "CdrTransactionDeleteEvent",
    "priority": "4",
    "timeOcurred": "2022-08-25T12:18:07.224Z",
    "title": "CdrTransactionDeleteEvent",
    "event": {
        "cDRTransaction": {
            "@type": "CdrTransaction",
            "href": "https://mycsp.com:8080/tmf-
api/CdrTransactionManagement/v5/CdrTransaction/1810",
```


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```
            "id": "1810",
            "relatedParty": [
                {
                    "@type": "RelatedPartyRefOrPartyRoleRef",
                    "role": "distributor",
                    "partyOrPartyRole": {
                      "@type": "PartyRef",
                      "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
                      "id": "2777",
                      "name": "MyFirstPartner",
                      "@referredType": "Organization"
                    }
                  },
                  {
                    "@type": "RelatedPartyRefOrPartyRoleRef",
                    "role": "revenueSharingModelOwner",
                    "partyOrPartyRole": {
                      "@type": "PartyRef",
                      "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2779",
                      "id": "2779",
                      "name": "Mobi Service Providor",
                      "@referredType": "Organization"
                    }
                  }
            ],
            "state": "charged",
            "requestedInitialState": "charged",
            "channel": [
                {
                    "@type": "RelatedChannel",
                    "role": "revenueSharingDepartment",
                    "channel": {
                        "@type": "ChannelRef",
                        "href": "https://mycsp.com:8080/tmf-
api/channelManagement/v5/channel/1777",
                        "id": "1777",
                        "name": "revenue sharing department",
                        "@referredType": "Channel"
                    }
                }
            ],
            "description": "the transaction of MyFirstPartner revenue share",
            "transactionItem": [
                {
                    "@type": "CdrTransactionItem",
                    "state": "billed",
                    "id": "18101",
                    "action": "add",
                    "cdr": [
                        {
                            "cdrDate": "2020-09-21T09:13:16-07:00",
                            "description": "Mobile handset 2023 distribution price",
                            "status": "charged",
                            "id": "181011",
                            "cdrType": "revenue share",
                            "relatedParty": {
                                "@type": "RelatedPartyRefOrPartyRoleRef",
                                "role": "distributor",
                                "partyOrPartyRole": {
                                    "@type": "PartyRef",
                                    "href": "https://mycsp.com:8080/tmf-
```


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```
api/partyManagement/v5/organization/2777",
                                    "id": "2777",
                                    "name": "MyFirstPartner",
                                    "@referredType": "Organization"
                                }
                            },
                            "isBilled": true,
                            "isTaxExempt": false,
                            "productRef": {
                                "@type": "ProductRef",
                                "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7810",
                                "id": "7810",
                                "name": "Mobile handset 2023 product of
MyFirstPartner",
                                "@referredType": "Product"
                            },
                            "taxExcludedRatingAmount": {
                                "unit": "USD",
                                "value": 0
                            },
                            "taxIncludedRatingAmount": {
                                "unit": "USD",
                                "value": 0
                            },
                            "taxRate": 0,
                            "amount": {
                                "unit": "USD",
                                "value": 50000
                            }
                        }
                    ],
                    "relatedParty": {
                        "@type": "RelatedPartyRefOrPartyRoleRef",
                        "role": "distributor",
                        "partyOrPartyRole": {
                            "@type": "PartyRef",
                            "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
                            "id": "2777",
                            "name": "MyFirstPartner",
                            "@referredType": "Organization"
                        }
                    },
                    "bill": [
                        {
                            "@type": "CustomerBillRef",
                            "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2001",
                            "id": "2001",
                            "name": "Customer 202302 bill of MyFirstPartner",
                            "@referredType": "CustomerBill"
                        }
                    ],
                    "payment": [
                        {
                            "@type": "PaymentRef",
                            "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/Payment/5001",
                            "id": "5001",
                            "name": "Customer 202302 bill payment of
MyFirstPartner",
                            "@referredType": "Payment"
```


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```
                        }
                    ]
                },
                {
                    "@type": "CdrTransactionItem",
                    "state": "billed",
                    "id": "18102",
                    "action": "add",
                    "cdr": [
                        {
                            "cdrDate": "2020-09-21T09:13:16-07:00",
                            "description": "smart speakers 2023 distribution price",
                            "status": "charged",
                            "id": "181021",
                            "cdrType": "revenue share",
                            "relatedParty": {
                                "@type": "RelatedPartyRefOrPartyRoleRef",
                                "role": "distributor",
                                "partyOrPartyRole": {
                                    "@type": "PartyRef",
                                    "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
                                    "id": "2777",
                                    "name": "MyFirstPartner",
                                    "@referredType": "Organization"
                                }
                            },
                            "isBilled": true,
                            "isTaxExempt": false,
                            "productRef": {
                                "@type": "ProductRef",
                                "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7811",
                                "id": "7811",
                                "name": "Smart speakers 2023 product of
MyFirstPartner",
                                "@referredType": "Product"
                            },
                            "taxExcludedRatingAmount": {
                                "unit": "USD",
                                "value": 0
                            },
                            "taxIncludedRatingAmount": {
                                "unit": "USD",
                                "value": 0
                            },
                            "taxRate": 0,
                            "amount": {
                                "unit": "USD",
                                "value": 30000
                            }
                        }
                    ],
                    "relatedParty": {
                        "@type": "RelatedPartyRefOrPartyRoleRef",
                        "role": "distributor",
                        "partyOrPartyRole": {
                            "@type": "PartyRef",
                            "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
                            "id": "2777",
                            "name": "MyFirstPartner",
                            "@referredType": "Organization"
```


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```
                        }
                    },
                    "bill": [
                        {
                            "@type": "CustomerBillRef",
                            "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2001",
                            "id": "2001",
                            "name": "Customer 202302 bill of MyFirstPartner",
                            "@referredType": "CustomerBill"
                        }
                    ],
                    "payment": [
                        {
                            "@type": "PaymentRef",
                            "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/Payment/5001",
                            "id": "5001",
                            "name": "Customer 202302 bill payment of
MyFirstPartner",
                            "@referredType": "Payment"
                        }
                    ]
                },
                {
                    "@type": "CdrTransactionItem",
                    "state": "billed",
                    "id": "18103",
                    "action": "add",
                    "cdr": [
                        {
                            "cdrDate": "2020-09-21T09:13:16-07:00",
                            "description": "Mobile handset 2023 distribution price",
                            "status": "charged",
                            "id": "181031",
                            "cdrType": "revenue share",
                            "relatedParty": {
                                "@type": "RelatedPartyRefOrPartyRoleRef",
                                "role": "revenueSharingModelOwner",
                                "partyOrPartyRole": {
                                    "@type": "PartyRef",
                                    "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2779",
                                    "id": "2779",
                                    "name": "Mobi Service Providor",
                                    "@referredType": "Organization"
                                }
                            },
                            "isBilled": true,
                            "isTaxExempt": false,
                            "productRef": {
                                "@type": "ProductRef",
                                "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7810",
                                "id": "7810",
                                "name": "Mobile handset 2023 product of
MyFirstPartner",
                                "@referredType": "Product"
                            },
                            "taxExcludedRatingAmount": {
                                "unit": "USD",
                                "value": 0
                            },
```


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```
                            "taxIncludedRatingAmount": {
                                "unit": "USD",
                                "value": 0
                            },
                            "taxRate": 0,
                            "amount": {
                                "unit": "USD",
                                "value": 60000
                            }
                        }
                    ],
                    "relatedParty": {
                        "@type": "RelatedPartyRefOrPartyRoleRef",
                        "role": "distributor",
                        "partyOrPartyRole": {
                            "@type": "PartyRef",
                            "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
                            "id": "2777",
                            "name": "MyFirstPartner",
                            "@referredType": "Organization"
                        }
                    },
                    "bill": [
                        {
                            "@type": "CustomerBillRef",
                            "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2002",
                            "id": "2002",
                            "name": "Customer 202302 bill of MyFirstPartner",
                            "@referredType": "CustomerBill"
                        }
                    ]
                }
            ],
            "creationDate": "2023-03-10T07:37:23.112Z",
            "completionDate": "2023-03-10T07:37:23.112Z"
        }
    },
    "reportingSystem": {
        "id": "759",
        "name": "APP-745",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "705",
        "name": "APP-317",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "CdrTransactionDeleteEvent"
}
```


Cdr Transaction State Change Event

Message example for CdrTransactionStateChangeEvent event


```
Content-Type: application/json
```


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```
{
    "correlationId": "95cfc73d-deb7",
    "description": "CdrTransactionStateChangeEvent illustration",
    "domain": "BusinessPartner",
    "eventId": "97d6-9751-40e4f01440c9",
    "eventTime": "2022-08-25T12:18:12.202Z",
    "eventType": "CdrTransactionStateChangeEvent",
    "priority": "4",
    "timeOcurred": "2022-08-25T12:18:07.224Z",
    "title": "CdrTransactionStateChangeEvent",
    "event": {
        "cDRTransaction": {
            "@type": "CdrTransaction",
            "@baseType": "CdrTransaction",
            "@schemaLocation": "https://mycsp.com:8080/tmf-
api/schemas/BusinessPartner/CdrTransaction.schema.json",
            "href": "https://mycsp.com:8080/tmf-
api/CdrTransactionManagement/v5/CdrTransaction/1810",
            "id": "1810",
            "relatedParty": [
                {
                    "@type": "RelatedPartyRefOrPartyRoleRef",
                    "role": "distributor",
                    "partyOrPartyRole": {
                      "@type": "PartyRef",
                      "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
                      "id": "2777",
                      "name": "MyFirstPartner",
                      "@referredType": "Organization"
                    }
                  },
                  {
                    "@type": "RelatedPartyRefOrPartyRoleRef",
                    "role": "revenueSharingModelOwner",
                    "partyOrPartyRole": {
                      "@type": "PartyRef",
                      "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2779",
                      "id": "2779",
                      "name": "Mobi Service Providor",
                      "@referredType": "Organization"
                    }
                  }
            ],
            "state": "charged",
            "requestedInitialState": "charged",
            "channel": [
                {
                "@type": "RelatedChannel",
                "@baseType": "RelatedChannel",
                "@schemaLocation": "https://mycsp.com:8080/tmf-
api/schemas/BusinessPartner/RelatedChannel.json",
                "role": "revenueSharingDepartment",
                "channel": {
                    "@type": "ChannelRef",
                    "href": "https://mycsp.com:8080/tmf-
api/channelManagement/v5/channel/1777",
                    "id": "1777",
                    "name": "revenue sharing department",
                    "@referredType": "Channel"
                }
```


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```
                }
            ],
            "description": "the transaction of MyFirstPartner revenue share",
            "transactionItem": [
                {
                    "@type": "CdrTransactionItem",
                    "state": "billed",
                    "id": "18101",
                    "action": "add",
                    "cdr": [
                        {
                            "cdrDate": "2020-09-21T09:13:16-07:00",
                            "description": "Mobile handset 2023 distribution price",
                            "status": "charged",
                            "id": "181011",
                            "cdrType": "revenue share",
                            "relatedParty": {
                                "@type": "RelatedPartyRefOrPartyRoleRef",
                                "role": "distributor",
                                "partyOrPartyRole": {
                                    "@type": "PartyRef",
                                    "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
                                    "id": "2777",
                                    "name": "MyFirstPartner",
                                    "@referredType": "Organization"
                                }
                            },
                            "isBilled": true,
                            "isTaxExempt": false,
                            "productRef": {
                                "@type": "ProductRef",
                                "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7810",
                                "id": "7810",
                                "name": "Mobile handset 2023 product of
MyFirstPartner",
                                "@referredType": "Product"
                            },
                            "taxExcludedRatingAmount": {
                                "unit": "USD",
                                "value": 0
                            },
                            "taxIncludedRatingAmount": {
                                "unit": "USD",
                                "value": 0
                            },
                            "taxRate": 0,
                            "amount": {
                                "unit": "USD",
                                "value": 50000
                            }
                        }
                    ],
                    "relatedParty": {
                        "@type": "RelatedPartyRefOrPartyRoleRef",
                        "role": "distributor",
                        "partyOrPartyRole": {
                            "@type": "PartyRef",
                            "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
                            "id": "2777",
                            "name": "MyFirstPartner",
```


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```
                            "@referredType": "Organization"
                        }
                    },
                    "bill": [
                        {
                            "@type": "CustomerBillRef",
                            "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2001",
                            "id": "2001",
                            "name": "Customer 202302 bill of MyFirstPartner",
                            "@referredType": "CustomerBill"
                        }
                    ],
                    "payment": [
                        {
                            "@type": "PaymentRef",
                            "id": "5001",
                            "name": "Customer 202302 bill payment of
MyFirstPartner",
                            "@referredType": "Payment"
                        }
                    ]
                },
                {
                    "@type": "CdrTransactionItem",
                    "state": "billed",
                    "id": "18102",
                    "action": "add",
                    "cdr": [
                        {
                            "cdrDate": "2020-09-21T09:13:16-07:00",
                            "description": "smart speakers 2023 distribution price",
                            "status": "charged",
                            "id": "181021",
                            "cdrType": "revenue share",
                            "relatedParty": {
                                "@type": "RelatedPartyRefOrPartyRoleRef",
                                "role": "distributor",
                                "partyOrPartyRole": {
                                    "@type": "PartyRef",
                                    "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
                                    "id": "2777",
                                    "name": "MyFirstPartner",
                                    "@referredType": "Organization"
                                }
                            },
                            "isBilled": true,
                            "isTaxExempt": false,
                            "productRef": {
                                "@type": "ProductRef",
                                "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7811",
                                "id": "7811",
                                "name": "Smart speakers 2023 product of
MyFirstPartner",
                                "@referredType": "Product"
                            },
                            "taxExcludedRatingAmount": {
                                "unit": "USD",
                                "value": 0
                            },
                            "taxIncludedRatingAmount": {
```


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```
                                "unit": "USD",
                                "value": 0
                            },
                            "taxRate": 0,
                            "amount": {
                                "unit": "USD",
                                "value": 30000
                            }
                        }
                    ],
                    "relatedParty": {
                        "@type": "RelatedPartyRefOrPartyRoleRef",
                        "role": "distributor",
                        "partyOrPartyRole": {
                            "@type": "PartyRef",
                            "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
                            "id": "2777",
                            "name": "MyFirstPartner",
                            "@referredType": "Organization"
                        }
                    },
                    "bill": [
                        {
                            "@type": "CustomerBillRef",
                            "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2001",
                            "id": "2001",
                            "name": "Customer 202302 bill of MyFirstPartner",
                            "@referredType": "CustomerBill"
                        }
                    ],
                    "payment": [
                        {
                            "@type": "PaymentRef",
                            "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/Payment/5001",
                            "id": "5001",
                            "name": "Customer 202302 bill payment of
MyFirstPartner",
                            "@referredType": "Payment"
                        }
                    ]
                },
                {
                    "@type": "CdrTransactionItem",
                    "state": "billed",
                    "id": "18102",
                    "action": "add",
                    "cdr": [
                            {
                            "cdrDate": "2020-09-21T09:13:16-07:00",
                            "description": "Mobile handset 2023 distribution price",
                            "status": "charged",
                            "id": "181031",
                            "cdrType": "revenue share",
                            "relatedParty": {
                                "@type": "RelatedPartyRefOrPartyRoleRef",
                                "role": "revenueSharingModelOwner",
                                "partyOrPartyRole": {
                                    "@type": "PartyRef",
                                    "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2779",
```


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```
                                    "id": "2779",
                                    "name": "Mobi Service Providor",
                                    "@referredType": "Organization"
                                }
                            },
                            "isBilled": true,
                            "isTaxExempt": false,
                            "productRef": {
                                "@type": "ProductRef",
                                "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7810",
                                "id": "7810",
                                "name": "Mobile handset 2023 product of
MyFirstPartner",
                                "@referredType": "Product"
                            },
                            "taxExcludedRatingAmount": {
                                "unit": "USD",
                                "value": 0
                            },
                            "taxIncludedRatingAmount": {
                                "unit": "USD",
                                "value": 0
                            },
                            "taxRate": 0,
                            "amount": {
                                "unit": "USD",
                                "value": 60000
                            }
                        }
                    ],
                    "relatedParty": {
                        "@type": "RelatedPartyRefOrPartyRoleRef",
                        "role": "distributor",
                        "partyOrPartyRole": {
                            "@type": "PartyRef",
                            "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
                            "id": "2777",
                            "name": "MyFirstPartner",
                            "@referredType": "Organization"
                        }
                    },
                    "bill": [
                        {
                            "@type": "CustomerBillRef",
                            "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2002",
                            "id": "2002",
                            "name": "Customer 202302 bill of MyFirstPartner",
                            "@referredType": "CustomerBill"
                        }
                    ]
                }
            ],
            "creationDate": "2023-03-10T07:37:23.112Z",
            "completionDate": "2023-03-10T07:37:23.112Z"
        }
    },
    "reportingSystem": {
        "id": "759",
        "name": "APP-745",
        "@type": "ReportingResource",
```


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```
            "@referredType": "LogicalResource"
        },
        "source": {
            "id": "705",
            "name": "APP-317",
            "@type": "ReportingResource",
            "@referredType": "LogicalResource"
        },
        "@baseType": "Event",
        "@type": "CdrTransactionStateChangeEvent"
    }
```


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API OPERATIONS

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

Operations on CdrTransaction

List or find CdrTransaction objects

`GET /cdrTransaction?fields=…`

Description

This operation list CdrTransaction entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving multiple cdr transactions.

Request

```
GET /cdrTransaction?relatedParty.partyOrPartyRole.id=2777
Content-Type: application/json
```

Response

```
200
```

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```
[
  {
    "@type": "CdrTransaction",
    "href": "https://mycsp.com:8080/tmf-
api/CdrTransactionManagement/v5/CdrTransaction/1810",
    "id": "1810",
    "relatedParty": [
      {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "distributor",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
          "id": "2777",
          "name": "MyFirstPartner",
          "@referredType": "Organization"
        }
      },
      {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "revenueSharingModelOwner",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2779",
          "id": "2779",
          "name": "Mobi Service Providor",
          "@referredType": "Organization"
        }
      }
    ],
    "state": "charged",
    "requestedInitialState": "charged",
    "channel": [
      {
        "@type": "RelatedChannel",
        "role": "revenueSharingDepartment",
        "channel": {
          "@type": "ChannelRef",
          "href": "https://mycsp.com:8080/tmf-
api/channelManagement/v5/channel/1777",
          "id": "1777",
          "name": "revenue sharing department",
          "@referredType": "Channel"
        }
      }
    ],
    "description": "the transaction of MyFirstPartner revenue share",
    "transactionItem": [
      {
        "@type": "CdrTransactionItem",
        "state": "billed",
        "id": "18101",
        "action": "add",
        "cdr": [
          {
            "cdrDate": "2020-09-21T09:13:16-07:00",
            "description": "Mobile handset 2023 distribution price",
            "status": "charged",
            "id": "181011",
            "cdrType": "revenue share",
            "relatedParty": {
```


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```
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "role": "distributor",
              "partyOrPartyRole": {
                "@type": "PartyRef",
                "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
                "id": "2777",
                "name": "MyFirstPartner",
                "@referredType": "Organization"
              }
            },
            "isBilled": true,
            "productRef": {
              "@type": "ProductRef",
              "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7810",
              "id": "7810",
              "name": "Mobile handset 2023 product of MyFirstPartner",
              "@referredType": "Product"
            },
            "taxExcludedRatingAmount": {
              "unit": "USD",
              "value": 0
            },
            "taxIncludedRatingAmount": {
              "unit": "USD",
              "value": 0
            },
            "taxRate": 0,
            "amount": {
              "unit": "USD",
              "value": 50000
            }
          }
        ],
        "relatedParty": {
          "@type": "RelatedPartyRefOrPartyRoleRef",
          "role": "distributor",
          "partyOrPartyRole": {
            "@type": "PartyRef",
            "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
            "id": "2777",
            "name": "MyFirstPartner",
            "@referredType": "Organization"
          }
        },
        "bill": [
          {
            "@type": "CustomerBillRef",
            "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2001",
            "id": "2001",
            "name": "Customer 202302 bill of MyFirstPartner",
            "@referredType": "CustomerBill"
          }
        ],
        "payment": [
          {
            "@type": "PaymentRef",
            "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/Payment/5001",
            "id": "5001",
```


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```
            "name": "Customer 202302 bill payment of MyFirstPartner",
            "@referredType": "Payment"
          }
        ]
      },
      {
        "@type": "CdrTransactionItem",
        "state": "billed",
        "id": "18102",
        "action": "add",
        "cdr": [
          {
            "cdrDate": "2020-09-21T09:13:16-07:00",
            "description": "smart speakers 2023 distribution price",
            "status": "charged",
            "id": "181021",
            "cdrType": "revenue share",
            "relatedParty": {
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "role": "distributor",
              "partyOrPartyRole": {
                "@type": "PartyRef",
                "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
                "id": "2777",
                "name": "MyFirstPartner",
                "@referredType": "Organization"
              }
            },
            "isBilled": true,
            "productRef": {
              "@type": "ProductRef",
              "id": "7811",
              "name": "Smart speakers 2023 product of MyFirstPartner",
              "@referredType": "Product"
            },
            "taxExcludedRatingAmount": {
              "unit": "USD",
              "value": 0
            },
            "taxIncludedRatingAmount": {
              "unit": "USD",
              "value": 0
            },
            "taxRate": 0,
            "amount": {
              "unit": "USD",
              "value": 30000
            }
          }
        ],
        "relatedParty": {
          "@type": "RelatedPartyRefOrPartyRoleRef",
          "role": "distributor",
          "partyOrPartyRole": {
            "@type": "PartyRef",
            "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
            "id": "2777",
            "name": "MyFirstPartner",
            "@referredType": "Organization"
          }
        },
```


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```
        "bill": [
          {
            "@type": "CustomerBillRef",
            "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2001",
            "id": "2001",
            "name": "Customer 202302 bill of MyFirstPartner",
            "@referredType": "CustomerBill"
          }
        ],
        "payment": [
          {
            "@type": "PaymentRef",
            "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/Payment/5001",
            "id": "5001",
            "name": "Customer 202302 bill payment of MyFirstPartner",
            "@referredType": "Payment"
          }
        ]
      },
      {
        "@type": "CdrTransactionItem",
        "state": "billed",
        "id": "18103",
        "action": "add",
        "cdr": [
          {
            "cdrDate": "2020-09-21T09:13:16-07:00",
            "description": "Mobile handset 2023 distribution price",
            "status": "charged",
            "id": "181031",
            "cdrType": "revenue share",
            "relatedParty": {
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "role": "revenueSharingModelOwner",
              "partyOrPartyRole": {
                "@type": "PartyRef",
                "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2779",
                "id": "2779",
                "name": "Mobi Service Providor",
                "@referredType": "Organization"
              }
            },
            "isBilled": true,
            "productRef": {
              "@type": "ProductRef",
              "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7810",
              "id": "7810",
              "name": "Mobile handset 2023 product of MyFirstPartner",
              "@referredType": "Product"
            },
            "taxExcludedRatingAmount": {
              "unit": "USD",
              "value": 0
            },
            "taxIncludedRatingAmount": {
              "unit": "USD",
              "value": 0
            },
            "taxRate": 0,
```


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```
            "amount": {
              "unit": "USD",
              "value": 60000
            }
          }
        ],
        "relatedParty": {
          "@type": "RelatedPartyRefOrPartyRoleRef",
          "role": "distributor",
          "partyOrPartyRole": {
            "@type": "PartyRef",
            "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
            "id": "2777",
            "name": "MyFirstPartner",
            "@referredType": "Organization"
          }
        },
        "bill": [
          {
            "@type": "CustomerBillRef",
            "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2002",
            "id": "2002",
            "name": "Customer 202302 bill of MyFirstPartner",
            "@referredType": "CustomerBill"
          }
        ]
      }
    ],
    "creationDate": "2023-03-10T07:37:23.112Z",
    "completionDate": "2023-03-10T07:37:23.112Z"
  }
]
```


Retrieves a CdrTransaction by ID

`GET /cdrTransaction/{id}?fields=…`

Description

This operation retrieves a CdrTransaction entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a specific cdr transaction.

Request

```
GET /cdrTransaction/1810
Content-Type: application/json
```

Response

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```
200

{
  "@type": "CdrTransaction",
  "href": "https://mycsp.com:8080/tmf-
api/CdrTransactionManagement/v5/CdrTransaction/1810",
  "id": "1810",
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "distributor",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
        "id": "2777",
        "name": "MyFirstPartner",
        "@referredType": "Organization"
      }
    },
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "revenueSharingModelOwner",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2779",
        "id": "2779",
        "name": "Mobi Service Providor",
        "@referredType": "Organization"
      }
    }
  ],
  "state": "charged",
  "requestedInitialState": "charged",
  "channel": [
    {
      "@type": "RelatedChannel",
      "role": "revenueSharingDepartment",
      "channel": {
        "@type": "ChannelRef",
        "href": "https://mycsp.com:8080/tmf-api/channelManagement/v5/channel/1777",
        "id": "1777",
        "name": "revenue sharing department",
        "@referredType": "Channel"
      }
    }
  ],
  "description": "the transaction of MyFirstPartner revenue share",
  "transactionItem": [
    {
      "@type": "CdrTransactionItem",
      "state": "billed",
      "id": "18101",
      "action": "add",
      "cdr": [
        {
          "cdrDate": "2020-09-21T09:13:16-07:00",
          "description": "Mobile handset 2023 distribution price",
          "status": "charged",
          "id": "181011",
          "cdrType": "revenue share",
```

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```
          "relatedParty": {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "distributor",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
              "id": "2777",
              "name": "MyFirstPartner",
              "@referredType": "Organization"
            }
          },
          "isBilled": true,
          "productRef": {
            "@type": "ProductRef",
            "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7810",
            "id": "7810",
            "name": "Mobile handset 2023 product of MyFirstPartner",
            "@referredType": "Product"
          },
          "taxExcludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxIncludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxRate": 0,
          "amount": {
            "unit": "USD",
            "value": 50000
          }
        }
      ],
      "relatedParty": {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "distributor",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
          "id": "2777",
          "name": "MyFirstPartner",
          "@referredType": "Organization"
        }
      },
      "bill": [
        {
          "@type": "CustomerBillRef",
          "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2001",
          "id": "2001",
          "name": "Customer 202302 bill of MyFirstPartner",
          "@referredType": "CustomerBill"
        }
      ],
      "payment": [
        {
          "@type": "PaymentRef",
          "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/Payment/5001",
```


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```
          "id": "5001",
          "name": "Customer 202302 bill payment of MyFirstPartner",
          "@referredType": "Payment"
        }
      ]
    },
    {
      "@type": "CdrTransactionItem",
      "state": "billed",
      "id": "18102",
      "action": "add",
      "cdr": [
        {
          "cdrDate": "2020-09-21T09:13:16-07:00",
          "description": "smart speakers 2023 distribution price",
          "status": "charged",
          "id": "181021",
          "cdrType": "revenue share",
          "relatedParty": {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "distributor",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
              "id": "2777",
              "name": "MyFirstPartner",
              "@referredType": "Organization"
            }
          },
          "isBilled": true,
          "productRef": {
            "@type": "ProductRef",
            "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7811",
            "id": "7811",
            "name": "Smart speakers 2023 product of MyFirstPartner",
            "@referredType": "Product"
          },
          "taxExcludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxIncludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxRate": 0,
          "amount": {
            "unit": "USD",
            "value": 30000
          }
        }
      ],
      "relatedParty": {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "distributor",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
          "id": "2777",
          "name": "MyFirstPartner",
```


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```
          "@referredType": "Organization"
        }
      },
      "bill": [
        {
          "@type": "CustomerBillRef",
          "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2001",
          "id": "2001",
          "name": "Customer 202302 bill of MyFirstPartner",
          "@referredType": "CustomerBill"
        }
      ],
      "payment": [
        {
          "@type": "PaymentRef",
          "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/Payment/5001",
          "id": "5001",
          "name": "Customer 202302 bill payment of MyFirstPartner",
          "@referredType": "Payment"
        }
      ]
    },
    {
      "@type": "CdrTransactionItem",
      "state": "billed",
      "id": "18103",
      "action": "add",
      "cdr": [
        {
          "cdrDate": "2020-09-21T09:13:16-07:00",
          "description": "Mobile handset 2023 distribution price",
          "status": "charged",
          "id": "181031",
          "cdrType": "revenue share",
          "relatedParty": {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "revenueSharingModelOwner",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2779",
              "id": "2779",
              "name": "Mobi Service Providor",
              "@referredType": "Organization"
            }
          },
          "isBilled": true,
          "productRef": {
            "@type": "ProductRef",
            "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7810",
            "id": "7810",
            "name": "Mobile handset 2023 product of MyFirstPartner",
            "@referredType": "Product"
          },
          "taxExcludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxIncludedRatingAmount": {
            "unit": "USD",
```


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```
            "value": 0
          },
          "taxRate": 0,
          "amount": {
            "unit": "USD",
            "value": 60000
          }
        }
      ],
      "relatedParty": {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "distributor",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
          "id": "2777",
          "name": "MyFirstPartner",
          "@referredType": "Organization"
        }
      },
      "bill": [
        {
          "@type": "CustomerBillRef",
          "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2002",
          "id": "2002",
          "name": "Customer 202302 bill of MyFirstPartner",
          "@referredType": "CustomerBill"
        }
      ]
    }
  ],
  "creationDate": "2023-03-10T07:37:23.112Z",
  "completionDate": "2023-03-10T07:37:23.112Z"
}
```


Creates a CdrTransaction

`POST /cdrTransaction?fields=…`

Description

This operation creates a CdrTransaction entity.

Mandatory Attributes

Mandatory Attributes	Rule
channel	
channel.@type	
channel.channel	
channel.channel.@type	
channel.channel.id	
channel.role	

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Mandatory Attributes	Rule
relatedParty	
relatedParty.@type	
relatedParty.partyOrPartyRole	
relatedParty.partyOrPartyRole.@type	
relatedParty.partyOrPartyRole.id	
relatedParty.role	
transactionItem	
transactionItem.@type	
transactionItem.action	
transactionItem.cdr	
transactionItem.cdr.cdrDirectionType	
transactionItem.cdr.isTaxExempt	
transactionItem.cdr.productRef	
transactionItem.cdr.productRef.@type	
transactionItem.cdr.productRef.id	
transactionItem.cdr.relatedParty	
transactionItem.cdr.relatedParty.@type	
transactionItem.cdr.relatedParty.partyOrPartyRole	
transactionItem.cdr.relatedParty.partyOrPartyRole.@type	
transactionItem.cdr.relatedParty.partyOrPartyRole.id	
transactionItem.cdr.relatedParty.role	
transactionItem.cdr.taxExcludedRatingAmount	
transactionItem.cdr.taxIncludedRatingAmount	
transactionItem.cdr.taxRate	
transactionItem.id	
transactionItem.relatedParty	
transactionItem.relatedParty.@type	

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Mandatory Attributes	Rule
transactionItem.relatedParty.partyOrPartyRole	
transactionItem.relatedParty.partyOrPartyRole.@type	
transactionItem.relatedParty.partyOrPartyRole.id	
transactionItem.relatedParty.role	
transactionItem.state	
@type	

Usage samples

Creation of a new CdrTransaction with POST operation

Request

```
POST /cdrTransaction
Content-Type: application/json

{
  "@type": "CdrTransaction",
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "distributor",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
        "id": "2777",
        "name": "MyFirstPartner",
        "@referredType": "Organization"
      }
    },
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "revenueSharingModelOwner",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2779",
        "id": "2779",
        "name": "Mobi Service Providor",
        "@referredType": "Organization"
      }
    }
  ],
  "state": "charged",
  "requestedInitialState": "charged",
  "channel": [
    {
      "@type": "RelatedChannel",
      "role": "revenueSharingDepartment",
      "channel": {
```

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```
        "@type": "ChannelRef",
        "href": "https://mycsp.com:8080/tmf-api/channelManagement/v5/channel/1777",
        "id": "1777",
        "name": "revenue sharing department",
        "@referredType": "Channel"
      }
    }
  ],
  "description": "the transaction of MyFirstPartner revenue share",
  "transactionItem": [
    {
      "@type": "CdrTransactionItem",
      "state": "billed",
      "action": "add",
      "cdr": [
        {
          "cdrDate": "2020-09-21T09:13:16-07:00",
          "description": "Mobile handset 2023 distribution price",
          "status": "charged",
          "cdrType": "revenue share",
          "relatedParty": {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "distributor",
            "partyOrPartyRole": {
                "@type": "PartyRef",
                "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
                "id": "2777",
                "name": "MyFirstPartner",
                "@referredType": "Organization"
            }
          },
          "isBilled": true,
          "isTaxExempt": false,
          "productRef": {
            "@type": "ProductRef",
            "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7810",
            "id": "7810",
            "name": "Mobile handset 2023 product of MyFirstPartner",
            "@referredType": "Product"
          },
          "taxExcludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxIncludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxRate": 0,
          "amount": {
            "unit": "USD",
            "value": 50000
          }
        }
      ],
      "relatedParty": {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "distributor",
        "partyOrPartyRole": {
            "@type": "PartyRef",
            "href": "https://mycsp.com:8080/tmf-
```


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```
api/partyManagement/v5/organization/2777",
            "id": "2777",
            "name": "MyFirstPartner",
            "@referredType": "Organization"
          }
        },
        "bill": [
          {
            "@type": "CustomerBillRef",
            "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2001",
            "id": "2001",
            "name": "Customer 202302 bill of MyFirstPartner",
            "@referredType": "CustomerBill"
          }
        ],
        "payment": [
          {
            "@type": "PaymentRef",
            "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/Payment/5001",
            "id": "5001",
            "name": "Customer 202302 bill payment of MyFirstPartner",
            "@referredType": "Payment"
          }
        ]
      },
      {
        "@type": "CdrTransactionItem",
        "state": "billed",
        "action": "add",
        "cdr": [
          {
            "cdrDate": "2020-09-21T09:13:16-07:00",
            "description": "smart speakers 2023 distribution price",
            "status": "charged",
            "cdrType": "revenue share",
            "relatedParty": {
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "role": "distributor",
              "partyOrPartyRole": {
                "@type": "PartyRef",
                "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
                "id": "2777",
                "name": "MyFirstPartner",
                "@referredType": "Organization"
              }
            },
            "isBilled": true,
            "isTaxExempt": false,
            "productRef": {
              "@type": "ProductRef",
              "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7811",
              "id": "7811",
              "name": "Smart speakers 2023 product of MyFirstPartner",
              "@referredType": "Product"
            },
            "taxExcludedRatingAmount": {
              "unit": "USD",
              "value": 0
            },
```


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```
          "taxIncludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxRate": 0,
          "amount": {
            "unit": "USD",
            "value": 30000
          }
        }
      ],
      "relatedParty": {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "distributor",
        "partyOrPartyRole": {
            "@type": "PartyRef",
            "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
            "id": "2777",
            "name": "MyFirstPartner",
            "@referredType": "Organization"
        }
      },
      "bill": [
        {
          "@type": "CustomerBillRef",
          "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2001",
          "id": "2001",
          "name": "Customer 202302 bill of MyFirstPartner",
          "@referredType": "CustomerBill"
        }
      ],
      "payment": [
        {
          "@type": "PaymentRef",
          "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/Payment/5001",
          "id": "5001",
          "name": "Customer 202302 bill payment of MyFirstPartner",
          "@referredType": "Payment"
        }
      ]
    },
    {
      "@type": "CdrTransactionItem",
      "state": "billed",
      "action": "add",
      "cdr": [
        {
          "cdrDate": "2020-09-21T09:13:16-07:00",
          "description": "Mobile handset 2023 distribution price",
          "status": "charged",
          "cdrType": "revenue share",
          "relatedParty": {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "revenueSharingModelOwner",
            "partyOrPartyRole": {
                "@type": "PartyRef",
                "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2779",
                "id": "2779",
                "name": "Mobi Service Provider",
```


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```
                "@referredType": "Organization"
              }
            },
            "isBilled": true,
            "isTaxExempt": false,
            "productRef": {
              "@type": "ProductRef",
              "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7810",
              "id": "7810",
              "name": "Mobile handset 2023 product of MyFirstPartner",
              "@referredType": "Product"
            },
            "taxExcludedRatingAmount": {
              "unit": "USD",
              "value": 0
            },
            "taxIncludedRatingAmount": {
              "unit": "USD",
              "value": 0
            },
            "taxRate": 0,
            "amount": {
              "unit": "USD",
              "value": 60000
            }
          }
        ],
        "relatedParty": {
          "@type": "RelatedPartyRefOrPartyRoleRef",
          "role": "distributor",
          "partyOrPartyRole": {
              "@type": "PartyRef",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
              "id": "2777",
              "name": "MyFirstPartner",
              "@referredType": "Organization"
          }
        },
        "bill": [
          {
            "@type": "CustomerBillRef",
            "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2002",
            "id": "2002",
            "name": "Customer 202302 bill of MyFirstPartner",
            "@referredType": "CustomerBill"
          }
        ]
      }
    ],
    "creationDate": "2023-03-10T07:37:23.112Z",
    "completionDate": "2023-03-10T07:37:23.112Z"
}
```


Response

```
201
```

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```
{
  "@type": "CdrTransaction",
  "href": "https://mycsp.com:8080/tmf-
api/CdrTransactionManagement/v5/CdrTransaction/1810",
  "id": "1810",
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "distributor",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
        "id": "2777",
        "name": "MyFirstPartner",
        "@referredType": "Organization"
      }
    },
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "revenueSharingModelOwner",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2779",
        "id": "2779",
        "name": "Mobi Service Providor",
        "@referredType": "Organization"
      }
    }
  ],
  "state": "charged",
  "requestedInitialState": "charged",
  "channel": [
    {
      "@type": "RelatedChannel",
      "role": "revenueSharingDepartment",
      "channel": {
        "@type": "ChannelRef",
        "href": "https://mycsp.com:8080/tmf-api/channelManagement/v5/channel/1777",
        "id": "1777",
        "name": "revenue sharing department",
        "@referredType": "Channel"
      }
    }
  ],
  "description": "the transaction of MyFirstPartner revenue share",
  "transactionItem": [
    {
      "@type": "CdrTransactionItem",
      "state": "billed",
      "id": "18101",
      "action": "add",
      "cdr": [
        {
          "cdrDate": "2020-09-21T09:13:16-07:00",
          "description": "Mobile handset 2023 distribution price",
          "status": "charged",
          "cdrType": "revenue share",
          "id": "181011",
          "relatedParty": {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "distributor",
```


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```
            "partyOrPartyRole": {
                "@type": "PartyRef",
                "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
                "id": "2777",
                "name": "MyFirstPartner",
                "@referredType": "Organization"
            }
          },
          "isBilled": true,
          "isTaxExempt": false,
          "productRef": {
            "@type": "ProductRef",
            "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7810",
            "id": "7810",
            "name": "Mobile handset 2023 product of MyFirstPartner",
            "@referredType": "Product"
          },
          "taxExcludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxIncludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxRate": 0,
          "amount": {
            "unit": "USD",
            "value": 50000
          }
        }
      ],
      "relatedParty": {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "distributor",
        "partyOrPartyRole": {
            "@type": "PartyRef",
            "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
            "id": "2777",
            "name": "MyFirstPartner",
            "@referredType": "Organization"
        }
      },
      "bill": [
        {
          "@type": "CustomerBillRef",
          "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2001",
          "id": "2001",
          "name": "Customer 202302 bill of MyFirstPartner",
          "@referredType": "CustomerBill"
        }
      ],
      "payment": [
        {
          "@type": "PaymentRef",
          "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/Payment/5001",
          "id": "5001",
          "name": "Customer 202302 bill payment of MyFirstPartner",
```


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```
          "@referredType": "Payment"
        }
      ]
    },
    {
      "@type": "CdrTransactionItem",
      "state": "billed",
      "id": "18102",
      "action": "add",
      "cdr": [
        {
          "cdrDate": "2020-09-21T09:13:16-07:00",
          "description": "smart speakers 2023 distribution price",
          "status": "charged",
          "id": "181021",
          "cdrType": "revenue share",
          "relatedParty": {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "distributor",
            "partyOrPartyRole": {
                "@type": "PartyRef",
                "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
                "id": "2777",
                "name": "MyFirstPartner",
                "@referredType": "Organization"
            }
          },
          "isBilled": true,
          "isTaxExempt": false,
          "productRef": {
            "@type": "ProductRef",
            "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7811",
            "id": "7811",
            "name": "Smart speakers 2023 product of MyFirstPartner",
            "@referredType": "Product"
          },
          "taxExcludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxIncludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxRate": 0,
          "amount": {
            "unit": "USD",
            "value": 30000
          }
        }
      ],
      "relatedParty": {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "distributor",
        "partyOrPartyRole": {
            "@type": "PartyRef",
            "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
            "id": "2777",
            "name": "MyFirstPartner",
            "@referredType": "Organization"
```


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```
        }
      },
      "bill": [
        {
          "@type": "CustomerBillRef",
          "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2001",
          "id": "2001",
          "name": "Customer 202302 bill of MyFirstPartner",
          "@referredType": "CustomerBill"
        }
      ],
      "payment": [
        {
          "@type": "PaymentRef",
          "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/Payment/5001",
          "id": "5001",
          "name": "Customer 202302 bill payment of MyFirstPartner",
          "@referredType": "Payment"
        }
      ]
    },
    {
      "@type": "CdrTransactionItem",
      "state": "billed",
      "id": "18103",
      "action": "add",
      "cdr": [
        {
          "cdrDate": "2020-09-21T09:13:16-07:00",
          "description": "Mobile handset 2023 distribution price",
          "status": "charged",
          "id": "181031",
          "cdrType": "revenue share",
          "relatedParty": {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "revenueSharingModelOwner",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2779",
              "id": "2779",
              "name": "Mobi Service Providor",
              "@referredType": "Organization"
            }
          },
          "isBilled": true,
          "isTaxExempt": false,
          "productRef": {
            "@type": "ProductRef",
            "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7810",
            "id": "7810",
            "name": "Mobile handset 2023 product of MyFirstPartner",
            "@referredType": "Product"
          },
          "taxExcludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxIncludedRatingAmount": {
            "unit": "USD",
```


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```
            "value": 0
          },
          "taxRate": 0,
          "amount": {
            "unit": "USD",
            "value": 60000
          }
        }
      ],
      "relatedParty": {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "distributor",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
          "id": "2777",
          "name": "MyFirstPartner",
          "@referredType": "Organization"
        }
      },
      "bill": [
        {
          "@type": "CustomerBillRef",
          "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2002",
          "id": "2002",
          "name": "Customer 202302 bill of MyFirstPartner",
          "@referredType": "CustomerBill"
        }
      ]
    }
  ],
  "creationDate": "2023-03-10T07:37:23.112Z",
  "completionDate": "2023-03-10T07:37:23.112Z"
}
```


Updates partially a CdrTransaction

`PATCH /cdrTransaction/{id}?fields=…`

Description

This operation allows partial updates of a CdrTransaction entity. Support of json/merge (https://tools.ietf.org/html/rfc7386) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optiona. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
href	
id	

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Non Patchable Attributes	Rule
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
cancellationDate	
cancellationReason	
channel	
completionDate	
creationDate	
description	
note	
relatedParty	
requestedInitialState	
state	
transactionItem	

Usage samples

PATCH CdrTransaction request leveraging json application example

Request

```
PATCH /cdrTransaction/{id}
Content-Type: application/json

{
  "@type": "CdrTransaction",
  "completionDate": "2023-05-10T07:37:23.112Z"
}
```

Response

```
200

{
  "@type": "CdrTransaction",
  "href": "https://mycsp.com:8080/tmf-
api/CdrTransactionManagement/v5/CdrTransaction/1810",
  "id": "1810",
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "distributor",
      "partyOrPartyRole": {
```

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```
        "@type": "PartyRef",
        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
        "id": "2777",
        "name": "MyFirstPartner",
        "@referredType": "Organization"
      }
    },
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "revenueSharingModelOwner",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2779",
        "id": "2779",
        "name": "Mobi Service Providor",
        "@referredType": "Organization"
      }
    }
  ],
  "state": "charged",
  "requestedInitialState": "charged",
  "channel": [
    {
      "@type": "RelatedChannel",
      "role": "revenueSharingDepartment",
      "channel": {
        "@type": "ChannelRef",
        "href": "https://mycsp.com:8080/tmf-api/channelManagement/v5/channel/1777",
        "id": "1777",
        "name": "revenue sharing department",
        "@referredType": "Channel"
      }
    }
  ],
  "description": "the transaction of MyFirstPartner revenue share",
  "transactionItem": [
    {
      "@type": "CdrTransactionItem",
      "state": "billed",
      "id": "18101",
      "action": "add",
      "cdr": [
        {
          "cdrDate": "2020-09-21T09:13:16-07:00",
          "description": "Mobile handset 2023 distribution price",
          "status": "charged",
          "id": "181011",
          "cdrType": "revenue share",
          "relatedParty": {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "distributor",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
              "id": "2777",
              "name": "MyFirstPartner",
              "@referredType": "Organization"
            }
          },
          "isBilled": true,
```


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```
          "productRef": {
            "@type": "ProductRef",
            "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7810",
            "id": "7810",
            "name": "Mobile handset 2023 product of MyFirstPartner",
            "@referredType": "Product"
          },
          "taxExcludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxIncludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxRate": 0,
          "amount": {
            "unit": "USD",
            "value": 50000
          }
        }
      ],
      "relatedParty": {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "distributor",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
          "id": "2777",
          "name": "MyFirstPartner",
          "@referredType": "Organization"
        }
      },
      "bill": [
        {
          "@type": "CustomerBillRef",
          "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2001",
          "id": "2001",
          "name": "Customer 202302 bill of MyFirstPartner",
          "@referredType": "CustomerBill"
        }
      ],
      "payment": [
        {
          "@type": "PaymentRef",
          "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/Payment/5001",
          "id": "5001",
          "name": "Customer 202302 bill payment of MyFirstPartner",
          "@referredType": "Payment"
        }
      ]
    },
    {
      "@type": "CdrTransactionItem",
      "state": "billed",
      "id": "18102",
      "action": "add",
      "cdr": [
        {
```


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```
          "cdrDate": "2020-09-21T09:13:16-07:00",
          "description": "smart speakers 2023 distribution price",
          "status": "charged",
          "id": "181021",
          "cdrType": "revenue share",
          "relatedParty": {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "distributor",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
              "id": "2777",
              "name": "MyFirstPartner",
              "@referredType": "Organization"
            }
          },
          "isBilled": true,
          "productRef": {
            "@type": "ProductRef",
            "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7811",
            "id": "7811",
            "name": "Smart speakers 2023 product of MyFirstPartner",
            "@referredType": "Product"
          },
          "taxExcludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxIncludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxRate": 0,
          "amount": {
            "unit": "USD",
            "value": 30000
          }
        }
      ],
      "relatedParty": {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "distributor",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
          "id": "2777",
          "name": "MyFirstPartner",
          "@referredType": "Organization"
        }
      },
      "bill": [
        {
          "@type": "CustomerBillRef",
           "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2001",
          "id": "2001",
          "name": "Customer 202302 bill of MyFirstPartner",
          "@referredType": "CustomerBill"
        }
      ],
```


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```
      "payment": [
        {
          "@type": "PaymentRef",
          "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/Payment/5001",
          "id": "5001",
          "name": "Customer 202302 bill payment of MyFirstPartner",
          "@referredType": "Payment"
        }
      ]
    },
    {
      "@type": "CdrTransactionItem",
      "state": "billed",
      "id": "18103",
      "action": "add",
      "cdr": [
        {
          "cdrDate": "2020-09-21T09:13:16-07:00",
          "description": "Mobile handset 2023 distribution price",
          "status": "charged",
          "id": "181031",
          "cdrType": "revenue share",
          "relatedParty": {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "revenueSharingModelOwner",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2779",
              "id": "2779",
              "name": "Mobi Service Providor",
              "@referredType": "Organization"
            }
          },
          "isBilled": true,
          "productRef": {
            "@type": "ProductRef",
            "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7810",
            "id": "7810",
            "name": "Mobile handset 2023 product of MyFirstPartner",
            "@referredType": "Product"
          },
          "taxExcludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxIncludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxRate": 0,
          "amount": {
            "unit": "USD",
            "value": 60000
          }
        }
      ],
      "relatedParty": {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "distributor",
        "partyOrPartyRole": {
```


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```
          "@type": "PartyRef",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
          "id": "2777",
          "name": "MyFirstPartner",
          "@referredType": "Organization"
        }
      },
      "bill": [
        {
          "@type": "CustomerBillRef",
          "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2002",
          "id": "2002",
          "name": "Customer 202302 bill of MyFirstPartner",
          "@referredType": "CustomerBill"
        }
      ]
    }
  ],
  "creationDate": "2023-03-10T07:37:23.112Z",
  "completionDate": "2023-05-10T07:37:23.112Z"
}
```


PATCH CdrTransaction request leveraging application merge-patch+json example

Request

```
PATCH /cdrTransaction/{id}
Content-Type: application/merge-patch+json

{
  "@type": "CdrTransaction",
  "completionDate": "2023-05-10T07:37:23.112Z"
}
```

Response


```
200

{
  "@type": "CdrTransaction",
  "href": "https://mycsp.com:8080/tmf-
api/CdrTransactionManagement/v5/CdrTransaction/1810",
  "id": "1810",
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "distributor",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
        "id": "2777",
        "name": "MyFirstPartner",
        "@referredType": "Organization"
      }
```


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```
    },
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "revenueSharingModelOwner",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2779",
        "id": "2779",
        "name": "Mobi Service Providor",
        "@referredType": "Organization"
      }
    }
  ],
  "state": "charged",
  "requestedInitialState": "charged",
  "channel": [
    {
      "@type": "RelatedChannel",
      "role": "revenueSharingDepartment",
      "channel": {
        "@type": "ChannelRef",
        "href": "https://mycsp.com:8080/tmf-api/channelManagement/v5/channel/1777",
        "id": "1777",
        "name": "revenue sharing department",
        "@referredType": "Channel"
      }
    }
  ],
  "description": "the transaction of MyFirstPartner revenue share",
  "transactionItem": [
    {
      "@type": "CdrTransactionItem",
      "state": "billed",
      "id": "18101",
      "action": "add",
      "cdr": [
        {
          "cdrDate": "2020-09-21T09:13:16-07:00",
          "description": "Mobile handset 2023 distribution price",
          "status": "charged",
          "id": "181011",
          "cdrType": "revenue share",
          "relatedParty": {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "distributor",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
              "id": "2777",
              "name": "MyFirstPartner",
              "@referredType": "Organization"
            }
          },
          "isBilled": true,
          "productRef": {
            "@type": "ProductRef",
            "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7810",
            "id": "7810",
            "name": "Mobile handset 2023 product of MyFirstPartner",
            "@referredType": "Product"
```


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```
          },
          "taxExcludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxIncludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxRate": 0,
          "amount": {
            "unit": "USD",
            "value": 50000
          }
        }
      ],
      "relatedParty": {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "distributor",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
          "id": "2777",
          "name": "MyFirstPartner",
          "@referredType": "Organization"
        }
      },
      "bill": [
        {
          "@type": "CustomerBillRef",
          "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2001",
          "id": "2001",
          "name": "Customer 202302 bill of MyFirstPartner",
          "@referredType": "CustomerBill"
        }
      ],
      "payment": [
        {
          "@type": "PaymentRef",
          "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/Payment/5001",
          "id": "5001",
          "name": "Customer 202302 bill payment of MyFirstPartner",
          "@referredType": "Payment"
        }
      ]
    },
    {
      "@type": "CdrTransactionItem",
      "state": "billed",
      "id": "18102",
      "action": "add",
      "cdr": [
        {
          "cdrDate": "2020-09-21T09:13:16-07:00",
          "description": "smart speakers 2023 distribution price",
          "status": "charged",
          "id": "181021",
          "cdrType": "revenue share",
          "relatedParty": {
            "@type": "RelatedPartyRefOrPartyRoleRef",
```


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```
            "role": "distributor",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
              "id": "2777",
              "name": "MyFirstPartner",
              "@referredType": "Organization"
            }
          },
          "isBilled": true,
          "productRef": {
            "@type": "ProductRef",
            "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7811",
            "id": "7811",
            "name": "Smart speakers 2023 product of MyFirstPartner",
            "@referredType": "Product"
          },
          "taxExcludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxIncludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxRate": 0,
          "amount": {
            "unit": "USD",
            "value": 30000
          }
        }
      ],
      "relatedParty": {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "distributor",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
          "id": "2777",
          "name": "MyFirstPartner",
          "@referredType": "Organization"
        }
      },
      "bill": [
        {
          "@type": "CustomerBillRef",
           "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2001",
          "id": "2001",
          "name": "Customer 202302 bill of MyFirstPartner",
          "@referredType": "CustomerBill"
        }
      ],
      "payment": [
        {
          "@type": "PaymentRef",
          "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/Payment/5001",
          "id": "5001",
          "name": "Customer 202302 bill payment of MyFirstPartner",
```


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```
          "@referredType": "Payment"
        }
      ]
    },
    {
      "@type": "CdrTransactionItem",
      "state": "billed",
      "id": "18103",
      "action": "add",
      "cdr": [
        {
          "cdrDate": "2020-09-21T09:13:16-07:00",
          "description": "Mobile handset 2023 distribution price",
          "status": "charged",
          "id": "181031",
          "cdrType": "revenue share",
          "relatedParty": {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "revenueSharingModelOwner",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2779",
              "id": "2779",
              "name": "Mobi Service Providor",
              "@referredType": "Organization"
            }
          },
          "isBilled": true,
          "productRef": {
            "@type": "ProductRef",
            "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7810",
            "id": "7810",
            "name": "Mobile handset 2023 product of MyFirstPartner",
            "@referredType": "Product"
          },
          "taxExcludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxIncludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxRate": 0,
          "amount": {
            "unit": "USD",
            "value": 60000
          }
        }
      ],
      "relatedParty": {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "distributor",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
          "id": "2777",
          "name": "MyFirstPartner",
          "@referredType": "Organization"
        }
```


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```
      },
      "bill": [
        {
          "@type": "CustomerBillRef",
          "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2002",
          "id": "2002",
          "name": "Customer 202302 bill of MyFirstPartner",
          "@referredType": "CustomerBill"
        }
      ]
    }
  ],
  "creationDate": "2023-03-10T07:37:23.112Z",
  "completionDate": "2023-05-10T07:37:23.112Z"
}
```


PATCH CdrTransaction request leveraging application json-patch+json example

Request

```
PATCH /cdrTransaction/{id}
Content-Type: application/json-patch+json

[
  {
    "op":"replace",
    "path": "/completionDate",
    "value": "2023-08-17T015:21:05.112Z"
  }
]
```

Response


```
200

{
  "@type": "CdrTransaction",
  "href": "https://mycsp.com:8080/tmf-
api/CdrTransactionManagement/v5/CdrTransaction/1810",
  "id": "1810",
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "distributor",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
        "id": "2777",
        "name": "MyFirstPartner",
        "@referredType": "Organization"
      }
    },
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "revenueSharingModelOwner",
```


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```
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2779",
        "id": "2779",
        "name": "Mobi Service Providor",
        "@referredType": "Organization"
      }
    }
  ],
  "state": "charged",
  "requestedInitialState": "charged",
  "channel": [
    {
      "@type": "RelatedChannel",
      "role": "revenueSharingDepartment",
      "channel": {
        "@type": "ChannelRef",
        "href": "https://mycsp.com:8080/tmf-api/channelManagement/v5/channel/1777",
        "id": "1777",
        "name": "revenue sharing department",
        "@referredType": "Channel"
      }
    }
  ],
  "description": "the transaction of MyFirstPartner revenue share",
  "transactionItem": [
    {
      "@type": "CdrTransactionItem",
      "state": "billed",
      "id": "18101",
      "action": "add",
      "cdr": [
        {
          "cdrDate": "2020-09-21T09:13:16-07:00",
          "description": "Mobile handset 2023 distribution price",
          "status": "charged",
          "id": "181011",
          "cdrType": "revenue share",
          "relatedParty": {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "distributor",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
              "id": "2777",
              "name": "MyFirstPartner",
              "@referredType": "Organization"
            }
          },
          "isBilled": true,
          "productRef": {
            "@type": "ProductRef",
            "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7810",
            "id": "7810",
            "name": "Mobile handset 2023 product of MyFirstPartner",
            "@referredType": "Product"
          },
          "taxExcludedRatingAmount": {
            "unit": "USD",
            "value": 0
```


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```
          },
          "taxIncludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxRate": 0,
          "amount": {
            "unit": "USD",
            "value": 50000
          }
        }
      ],
      "relatedParty": {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "distributor",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
          "id": "2777",
          "name": "MyFirstPartner",
          "@referredType": "Organization"
        }
      },
      "bill": [
        {
          "@type": "CustomerBillRef",
          "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2001",
          "id": "2001",
          "name": "Customer 202302 bill of MyFirstPartner",
          "@referredType": "CustomerBill"
        }
      ],
      "payment": [
        {
          "@type": "PaymentRef",
          "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/Payment/5001",
          "id": "5001",
          "name": "Customer 202302 bill payment of MyFirstPartner",
          "@referredType": "Payment"
        }
      ]
    },
    {
      "@type": "CdrTransactionItem",
      "state": "billed",
      "id": "18102",
      "action": "add",
      "cdr": [
        {
          "cdrDate": "2020-09-21T09:13:16-07:00",
          "description": "smart speakers 2023 distribution price",
          "status": "charged",
          "id": "181021",
          "cdrType": "revenue share",
          "relatedParty": {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "distributor",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "href": "https://mycsp.com:8080/tmf-
```


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```
api/partyManagement/v5/organization/2777",
              "id": "2777",
              "name": "MyFirstPartner",
              "@referredType": "Organization"
            }
          },
          "isBilled": true,
          "productRef": {
            "@type": "ProductRef",
            "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7811",
            "id": "7811",
            "name": "Smart speakers 2023 product of MyFirstPartner",
            "@referredType": "Product"
          },
          "taxExcludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxIncludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxRate": 0,
          "amount": {
            "unit": "USD",
            "value": 30000
          }
        }
      ],
      "relatedParty": {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "distributor",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
          "id": "2777",
          "name": "MyFirstPartner",
          "@referredType": "Organization"
        }
      },
      "bill": [
        {
          "@type": "CustomerBillRef",
          "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2001",
          "id": "2001",
          "name": "Customer 202302 bill of MyFirstPartner",
          "@referredType": "CustomerBill"
        }
      ],
      "payment": [
        {
          "@type": "PaymentRef",
          "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/Payment/5001",
          "id": "5001",
          "name": "Customer 202302 bill payment of MyFirstPartner",
          "@referredType": "Payment"
        }
      ]
    },
```


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```
    {
      "@type": "CdrTransactionItem",
      "state": "billed",
      "id": "18103",
      "action": "add",
      "cdr": [
        {
          "cdrDate": "2020-09-21T09:13:16-07:00",
          "description": "Mobile handset 2023 distribution price",
          "status": "charged",
          "id": "181031",
          "cdrType": "revenue share",
          "relatedParty": {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "revenueSharingModelOwner",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2779",
              "id": "2779",
              "name": "Mobi Service Providor",
              "@referredType": "Organization"
            }
          },
          "isBilled": true,
          "productRef": {
            "@type": "ProductRef",
            "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7810",
            "id": "7810",
            "name": "Mobile handset 2023 product of MyFirstPartner",
            "@referredType": "Product"
          },
          "taxExcludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxIncludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxRate": 0,
          "amount": {
            "unit": "USD",
            "value": 60000
          }
        }
      ],
      "relatedParty": {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "distributor",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
          "id": "2777",
          "name": "MyFirstPartner",
          "@referredType": "Organization"
        }
      },
      "bill": [
        {
          "@type": "CustomerBillRef",
```


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```
          "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2002",
          "id": "2002",
          "name": "Customer 202302 bill of MyFirstPartner",
          "@referredType": "CustomerBill"
        }
      ]
    }
  ],
  "creationDate": "2023-03-10T07:37:23.112Z",
  "completionDate": "2023-08-17T015:21:05.112Z"
}
```

PATCH CdrTransaction request leveraging application json-patch+json example

Request

```
PATCH /cdrTransaction/{id}
Content-Type: application/json-patch-query+json

[
  {
    "op":"replace",
    "path": "/transactionItem.state?id=18101",
    "value": "completed"
  }
]
```

Response

```
200

{
  "@type": "CdrTransaction",
  "href": "https://mycsp.com:8080/tmf-
api/CdrTransactionManagement/v5/CdrTransaction/1810",
  "id": "1810",
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "distributor",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
        "id": "2777",
        "name": "MyFirstPartner",
        "@referredType": "Organization"
      }
    },
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "revenueSharingModelOwner",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2779",
```

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```
        "id": "2779",
        "name": "Mobi Service Providor",
        "@referredType": "Organization"
      }
    }
  ],
  "state": "charged",
  "requestedInitialState": "charged",
  "channel": [
    {
      "@type": "RelatedChannel",
      "role": "revenueSharingDepartment",
      "channel": {
        "@type": "ChannelRef",
        "href": "https://mycsp.com:8080/tmf-api/channelManagement/v5/channel/1777",
        "id": "1777",
        "name": "revenue sharing department",
        "@referredType": "Channel"
      }
    }
  ],
  "description": "the transaction of MyFirstPartner revenue share",
  "transactionItem": [
    {
      "@type": "CdrTransactionItem",
      "state": "completed",
      "id": "18101",
      "action": "add",
      "cdr": [
        {
          "cdrDate": "2020-09-21T09:13:16-07:00",
          "description": "Mobile handset 2023 distribution price",
          "status": "charged",
          "id": "181011",
          "cdrType": "revenue share",
          "relatedParty": {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "distributor",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
              "id": "2777",
              "name": "MyFirstPartner",
              "@referredType": "Organization"
            }
          },
          "isBilled": true,
          "productRef": {
            "@type": "ProductRef",
            "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7810",
            "id": "7810",
            "name": "Mobile handset 2023 product of MyFirstPartner",
            "@referredType": "Product"
          },
          "taxExcludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxIncludedRatingAmount": {
            "unit": "USD",
            "value": 0
```


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```
          },
          "taxRate": 0,
          "amount": {
            "unit": "USD",
            "value": 50000
          }
        }
      ],
      "relatedParty": {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "distributor",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
          "id": "2777",
          "name": "MyFirstPartner",
          "@referredType": "Organization"
        }
      },
      "bill": [
        {
          "@type": "CustomerBillRef",
          "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2001",
          "id": "2001",
          "name": "Customer 202302 bill of MyFirstPartner",
          "@referredType": "CustomerBill"
        }
      ],
      "payment": [
        {
          "@type": "PaymentRef",
          "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/Payment/5001",
          "id": "5001",
          "name": "Customer 202302 bill payment of MyFirstPartner",
          "@referredType": "Payment"
        }
      ]
    },
    {
      "@type": "CdrTransactionItem",
      "state": "billed",
      "id": "18102",
      "action": "add",
      "cdr": [
        {
          "cdrDate": "2020-09-21T09:13:16-07:00",
          "description": "smart speakers 2023 distribution price",
          "status": "charged",
          "id": "181021",
          "cdrType": "revenue share",
          "relatedParty": {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "distributor",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
              "id": "2777",
              "name": "MyFirstPartner",
              "@referredType": "Organization"
```


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```
            }
          },
          "isBilled": true,
          "productRef": {
            "@type": "ProductRef",
            "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7811",
            "id": "7811",
            "name": "Smart speakers 2023 product of MyFirstPartner",
            "@referredType": "Product"
          },
          "taxExcludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxIncludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxRate": 0,
          "amount": {
            "unit": "USD",
            "value": 30000
          }
        }
      ],
      "relatedParty": {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "distributor",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
          "id": "2777",
          "name": "MyFirstPartner",
          "@referredType": "Organization"
        }
      },
      "bill": [
        {
          "@type": "CustomerBillRef",
          "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2001",
          "id": "2001",
          "name": "Customer 202302 bill of MyFirstPartner",
          "@referredType": "CustomerBill"
        }
      ],
      "payment": [
        {
          "@type": "PaymentRef",
          "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/Payment/5001",
          "id": "5001",
          "name": "Customer 202302 bill payment of MyFirstPartner",
          "@referredType": "Payment"
        }
      ]
    },
    {
      "@type": "CdrTransactionItem",
      "state": "billed",
      "id": "18103",
```


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```
      "action": "add",
      "cdr": [
        {
          "cdrDate": "2020-09-21T09:13:16-07:00",
          "description": "Mobile handset 2023 distribution price",
          "status": "charged",
          "id": "181031",
          "cdrType": "revenue share",
          "relatedParty": {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "revenueSharingModelOwner",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2779",
              "id": "2779",
              "name": "Mobi Service Providor",
              "@referredType": "Organization"
            }
          },
          "isBilled": true,
          "productRef": {
            "@type": "ProductRef",
            "href": "https://mycsp.com:8080/tmf-
api/ProductManagement/v5/Product/7810",
            "id": "7810",
            "name": "Mobile handset 2023 product of MyFirstPartner",
            "@referredType": "Product"
          },
          "taxExcludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxIncludedRatingAmount": {
            "unit": "USD",
            "value": 0
          },
          "taxRate": 0,
          "amount": {
            "unit": "USD",
            "value": 60000
          }
        }
      ],
      "relatedParty": {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "distributor",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/2777",
          "id": "2777",
          "name": "MyFirstPartner",
          "@referredType": "Organization"
        }
      },
      "bill": [
        {
          "@type": "CustomerBillRef",
          "href": "https://mycsp.com:8080/tmf-
api/CustomerManagement/v5/CustomerBill/2002",
          "id": "2002",
          "name": "Customer 202302 bill of MyFirstPartner",
```


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```
            "@referredType": "CustomerBill"
          }
        ]
      }
    ],
    "creationDate": "2023-03-10T07:37:23.112Z",
    "completionDate": "2023-05-10T07:37:23.112Z"
  }
```


Deletes a CdrTransaction

`DELETE /cdrTransaction/{id}`

Description

This operation deletes a CdrTransaction entity.

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API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

Register listener

POST /hub

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

Usage Samples

Here's an example of a request for registering a listener.

Request

```
POST /api/hub

Accept: application/json

{
    "callback": "http://in.listener.com"
}
```

Response

```
201

Content-Type: application/json

Location: /api/hub/42

{
    "id":"42",
    "callback": "http://in.listener.com",
    "query":""
}
```

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Unregister listener

DELETE /hub/{id}

Description

Clears the communication endpoint address that was set by creating the Hub..

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request

```
DELETE /api/hub/42

Accept: application/json
```

Response

```
204
```

Publish Event to listener

POST /client/listener

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

Usage Samples

Here's an example of a notification received by the listener. In this example "EVENT TYPE" should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

Request

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```
POST /client/listener

Accept: application/json

{
    "event": {
        EVENT BODY
    },
    "eventType": "EVENT_TYPE"
}
```

Response

```
201
```

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

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Acknowledgements

Release History

Release Number	Date	Release led by:	Description
Version: 5.0.0	August 2023	Lianhua Zhang AsiaInfo zhanglh2@asiainfo.com Yun Li AsiaInfo liyun9@asiainfo.com Pierre Gauthier TM Forum pgauthier@tmforum.org Henrique Rodrigues TM Forum hrodrigues@tmforum.org	First Release of the Document.

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